IMAGE PROCESSING · LOCAL OPERATIONS

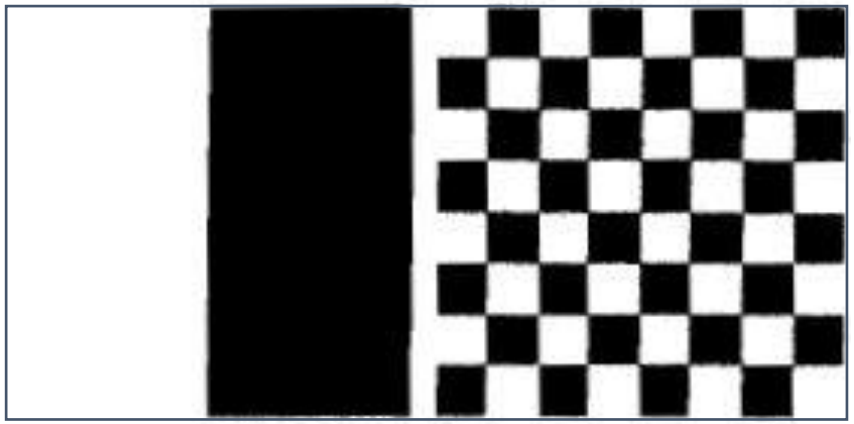
Local histogram processing

How neighborhoods, filters, and derivatives turn raw pixels into clearer images.

Example

1) Laplacian with -8 in the center yields sharper result than one with a -4 in the center explain reason in detail
2) Suppose that each image is blurred with a 3*3 averaging mask
   a) The histogram of the following images are the same?
   b) Would the histograms of blurred images equal?

   a& b → explain and sketch



Example solution

1) Laplacian with -8 in the center yields sharper result than one with a -4 in the center

   explain reason in detail

1	1	1
1	-8	1
1	1	1

0	1	0
1	-4	1
0	1	0

0	1	0
0	-2	0
0	1	0

0	0	0
1	-2	1
0	0	0

A Laplacian mask with **-8 at the center** produces a sharper result than one with -4 at the center because it detects intensity changes in more directions.

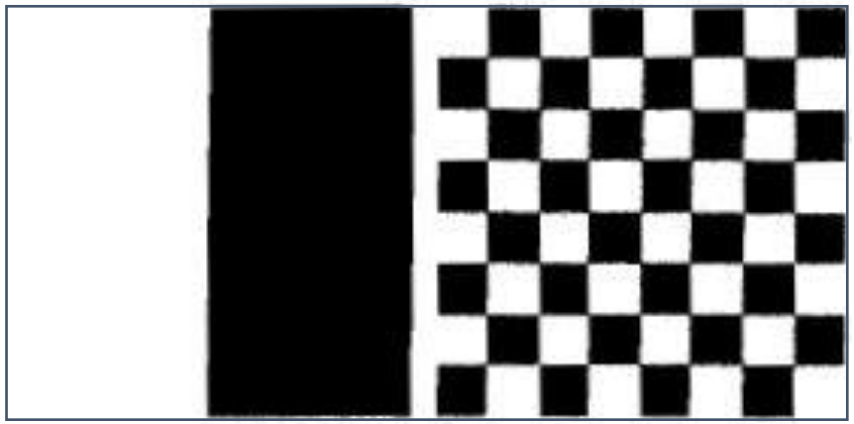
- **-2 Laplacian mask:** Detects intensity changes in only one direction (horizontal or vertical), so it sharpens the image in one direction only.
- **-4 Laplacian mask:** Detects intensity changes in the horizontal and vertical directions only.
- **-8 Laplacian mask:** Detects changes in the horizontal, vertical, and diagonal directions.

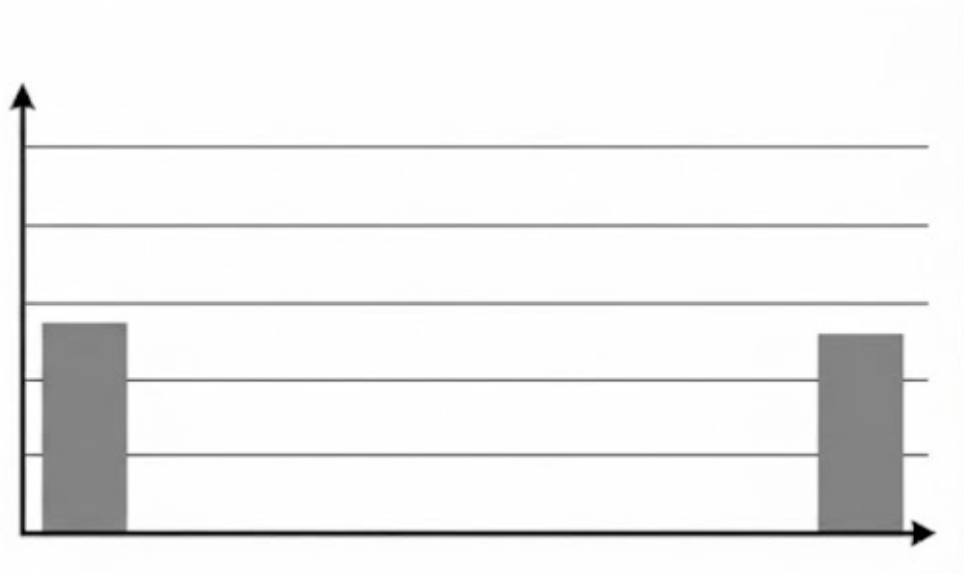
Therefore, the -8 mask detects more edges and fine details, resulting in stronger edge enhancement and a generally sharper-looking image.

Example solution

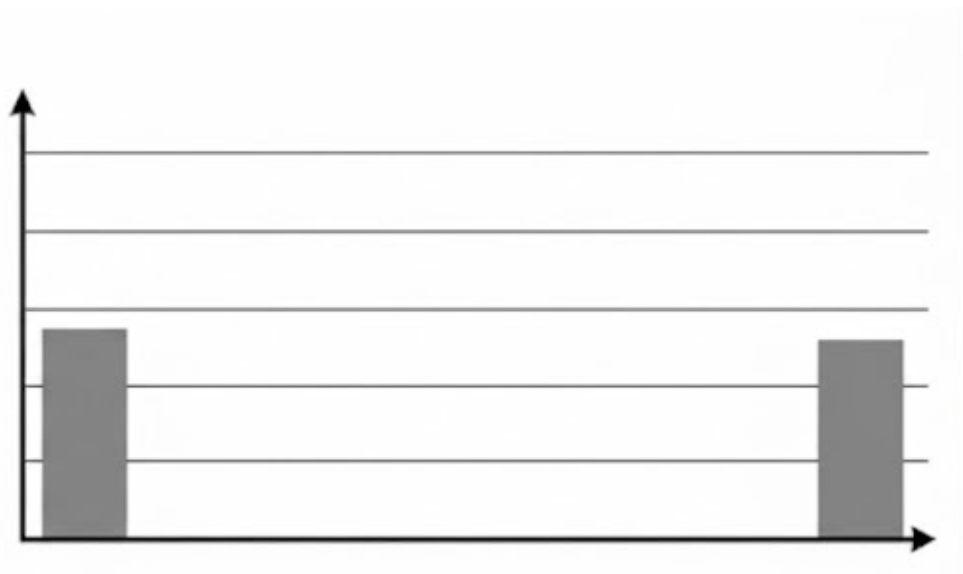
2) Suppose that each image is blurred with a 3*3 averaging mask

   a) The histogram of the following images are the same? **Yes**





Left image



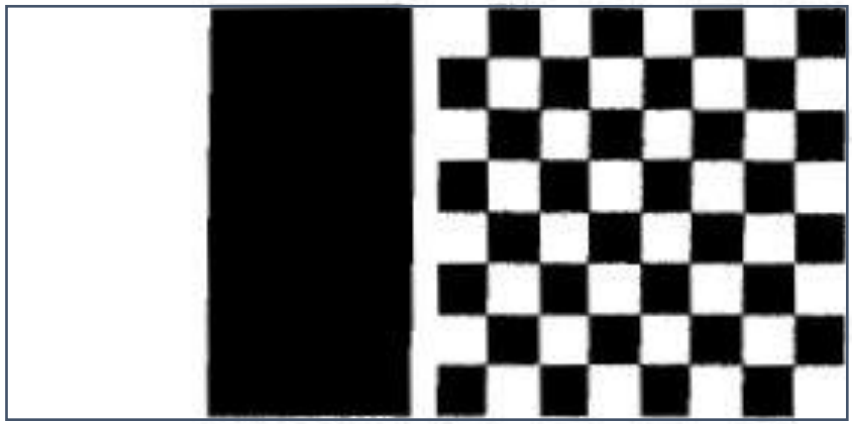
Right image

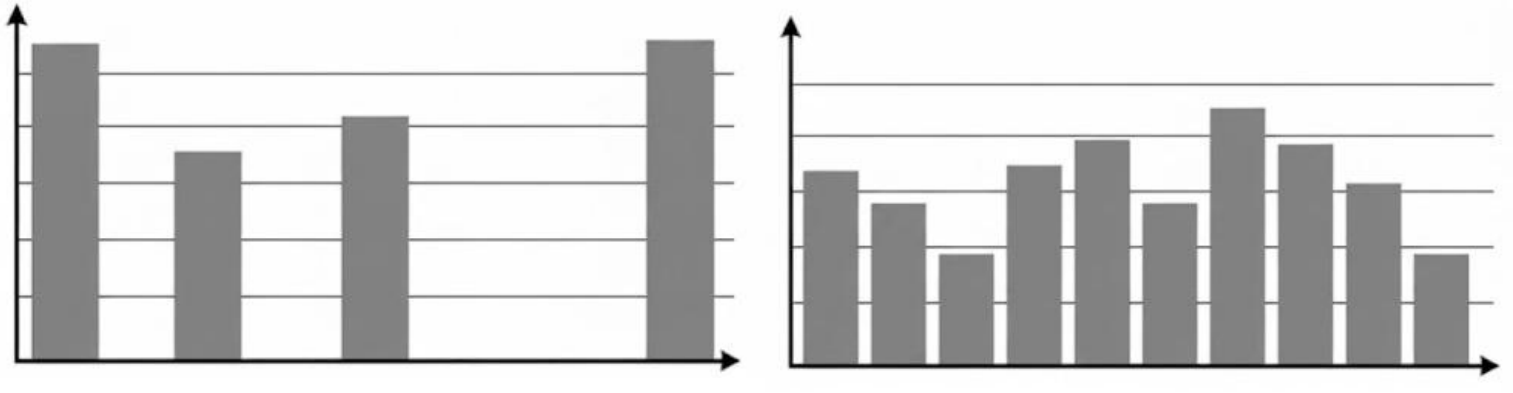
Example solution

2) Suppose that each image is blurred with a 3*3 averaging mask

   b) Would the histograms of blurred images equal? **No**

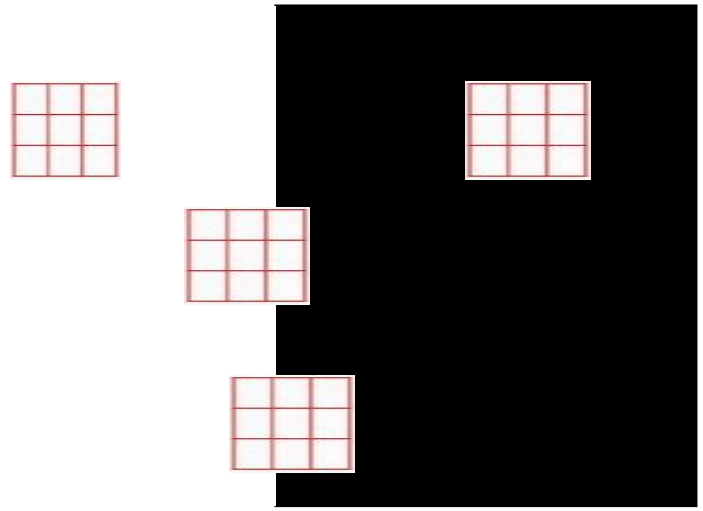
a& b → explain and sketch





Left image

Right image



Example cont.

3) Using the figure to the right, find the filtered image using zero padding of the original image

2	9	10	0
7	1	6	1
10	15	2	6
11	3	8	10

Original image

1/4

0	1	0
1	0	1
0	1	0

Shift – invariant
Linear filter

?	?	?	?
?	?	?	?
?	?	?	?
?	?	?	?

Filtered image

Example solution

3) Using the figure to the right, find the filtered image using zero padding of the original image

2	9	10	0
7	1	6	1
10	15	2	6
11	3	8	10

Original image

1/4

0	1	0
1	0	1
0	1	0

Shift – invariant Linear filter

0	0	0	0	0	0
0	2	9	10	0	0
0	7	1	6	1	0
0	10	15	2	6	0
0	11	3	8	10	0
0	0	0	0	0	0

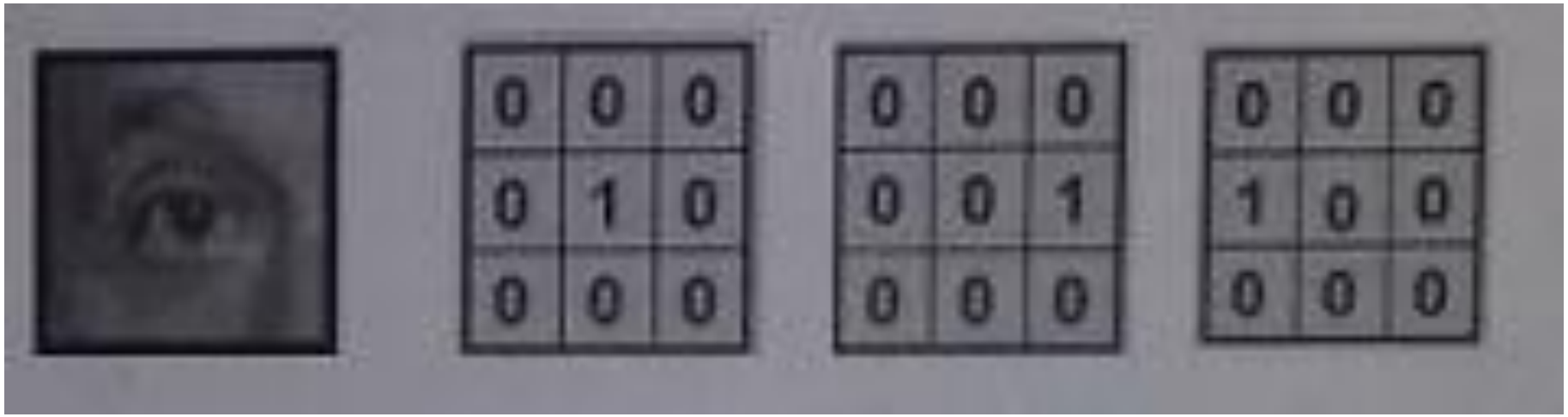
Padded image

4	3	4	3
3	9	4	3
8	4	9	3
3	9	4	4

Filtered image

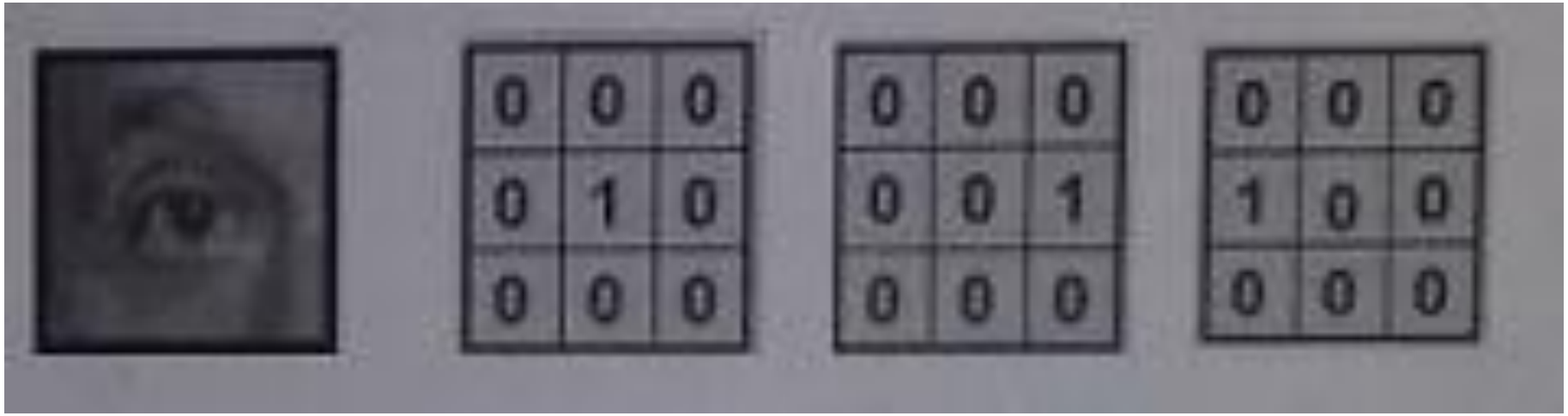
Example cont.

4) Predict the output using the following correlation filters for the image in the following figure?



Example solution

4) Predict the output using the following correlation filters for the image in the following figure?



No change output image is same as input image

Input image is shifted to left by one pixel

Input image is shifted to right by one pixel

Linear and non-linear filters answer different questions

LINEAR

What weighted average represents this neighborhood?

Each neighbor contributes through a fixed coefficient. The same arithmetic structure applies whether the window is smooth, noisy, or crossing an edge.

NON-LINEAR

Which value, rank, condition, or shape best represents it?

Outliers and boundaries can change the decision. The rule adapts to the neighborhood rather than preserving fixed addition.

Trade-off: stronger control over outliers and edges, less convenient frequency-domain analysis.

THE Non-linear filter RULE

The Non-linear filter turns sorting into denoising

The non-linear filter suppresses isolated extremes while keeping many boundaries sharper than a mean filter.

01

Collect

Collect the gray levels in a local window.

02

Sort

Sort the values from low to high.

03

Replace

Replace the center pixel with the middle value.

The operation is local, but the decision depends on rank rather than weighted addition.

Rank filters choose a position in the local histogram

Rank filters use local gray-level ordering. The rule chooses a position, then returns the value occupying that position.

LOWEST RANK

`min`

Minimum

Select the lowest gray level.

MIDDLE RANK

`median`

Median

Select the middle gray level.

HIGHEST RANK

`max`

Maximum

Select the highest gray level.

Midpoint RANK

`midpoint`

Select the Midpoint filter [(min+max)/2].

A 3×3 median filter example makes the rule visible

One bright impulse sits inside an otherwise quiet neighborhood. The sorted list makes the replacement unambiguous.

Neighborhood	Sorted values	Output
12, 13, 14 / 13, 255, 14 / 12, 14, 13	12, 12, 13, 13, **13**, 14, 14, 14, 255	13

CENTER PIXEL

255

before

take the middle-ordered value

FILTERED OUTPUT

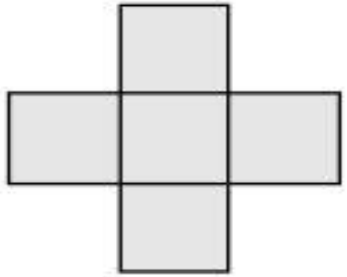
13

after

A 3×3 median filter example makes the rule visible

Filter the image using a Median filter (after replicate-padding), where the filter mask as given in figure.

7	1	7	1
5	3	5	3
7	1	7	1
5	3	5	3



1 5 7 7 7

7	7	1	7	1	1
7	7	1	7	1	1
5	5	3	5	3	3
7	7	1	7	1	1
5	5	3	5	3	3
5	5	3	5	3	3

Output image

7	3	5	1
5	3	5	3
5	3	5	3
5	3	5	3

Think about it!

Is the Mean Filter a separable filter?

$$\frac{1}{9}\begin{bmatrix}1 & 1 & 1\\ 1 & 1 & 1\\ 1 & 1 & 1\end{bmatrix} = \frac{1}{3}\begin{bmatrix}1\\ 1\\ 1\end{bmatrix} \frac{1}{3}\begin{bmatrix}1 & 1 & 1\end{bmatrix}$$

Think about it!

Is separable filter?

1	-2	1
-2	4	-2
1	-2	1

=

1
-2
1

1	-2	1